The artifact I selected is an android inventory management app that I created for my CS 360 project. It basically allows users to add, delete, update, and view inventory items. The old version used SQLite via the DatabaseHelper.java class to handle practically everything. For this enhancement, I focused on shifting away from SQLite to android Room and using Room's CRUD functions instead of my own.

I selected this artifact because it allows me to demonstrate my skills in database design and layered architecture in an android application. To summarize, this enhancement introduced a Room @Entity through the InvnetoryItem data class, an InventoryDao interface for database operations, and the InventoryDatabase class, which acts as the new functioning Room database.

This enhancement also reinforced the other enhancements the most since it involved wiring-up ActivityMain and InventoryViewModel with CRUD functionality and the Room database. So, DatabaseHelper now only handles ActivityLogin user table, which still uses SQLite.

Enhancement three supports the course outcome of demonstrating the ability to use well-founded techniques and tools in computing practices to implement computer solutions that deliver value and accomplish industry-specific goals. It also supports the course outcome of designing and evaluating computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution, while managing the trade-offs involved in design choices. Room is the more appropriate, standard practice for most android apps, and switching databases involves changing and upgrading current architecture and structure.

A big challenge of mine I encountered while finishing this enhancement would be integrating Room without disrupting the functionality from earlier enhancements. I also found articles online that suggested changing code from previous enhancement for the sake of efficiency. Specifically, using async database operations, which involves creating new functions, editing current functions, and disrupting the established flow I previously created via the UML class diagrams. Managing project scope has also been an ongoing challenge. I learned to stay on track and not get lost in the possibilities of improvement. Basically, if the project is not fundamentally finished, looking for new features is pointless.